

Hello.

Hi.

Hello.

Manor Medical. There is a doctor with us, Tahover.

Shalom.

Hi. Manishmastas. Manishmastas.

We can in English. For convenience of everyone.

Yes, yes, yes, yes, great.

Okay.

So, who is Anastasia? Your mother or your...

My sister. She's 47 years old.

Ah, okay. Okay. And where is she now?

What do you mean?

Is she with you now?

No, she's not with me. She is recovering. Currently at her place. She went at her home. After treatment. So, she's living in a house near the city. And she's right now recovering there. I talked to her. She said at first she felt very well. And she had a very big expansion of the liver during this year. Like she has had a stomach increased significantly. Right now she says that it's all gone. And the belly is flat already. But she feels... In the second part of the day she feels very tired and weak. And she says she feels intoxication. Very strong. And also the doctor ordered that she should walk a lot. And she said that she needs to walk a lot. Because otherwise she cannot get up. So... And she feels bad. But she says right now that when she walks she has her head... I don't know. A fuse. I don't know how to say that. So, let me translate. Maybe the translator can translate.

Dizziness.

Yes, and she doesn't know why that. Is that due to medication or maybe due to the PRES she'd experienced half of the year ago, Pazopanib or something else. So that's her condition and she wonders if her condition ever becomes better?

Okay, so let's go over everything you said. Because it's a complicated case. So, I see here. According to what they put in the file. So, this

is the PET CT from two months ago. Where they saw the liver metastasis. It grew a bit. Also between the stomach and the pancreas. And the uterus didn't have any uptake. So mainly in the left PET the concern is mainly the liver. Then we have the pathology. The revision of pathology. Just from a few days ago. The biopsy from the abdominal cavity and the liver. And it shows a gist. The MRI of the abdomen. I'm trying to understand. They thought it was a leiomyosarcoma. And then they decided it was a gist or did she have both?

At the very beginning she was diagnosed that this is a leiomyosarcoma. The National Institute of Cancer in Kyiv was 100% confident that this is a leiomyosarcoma. So they didn't do immunohistochemistry. Because they took insufficient material on biopsy. And also they feared that they will cause strong metastasis. If they pin the tissue. So they were afraid that the tumor will metastasize if they take the biopsy again. And they diagnosed it as a leiomyosarcoma. They said that they are 100% confident. And during the year she received treatment from leiomyosarcoma. This was first line and second line chemotherapy. And after we did the PET CT. We found that the uterus and ovary are metabolically inactive. But they were active on the first CT as far as I understand. They accumulated contrasts on the very first CT which was made on August. And after that there was a suggestion made that it might be something else. In liver and near the stomach. And after that the new biopsy was made. And it suggested that this is a GIST.

So let's see. Because it's still not very clear. They think she had both? Two kinds of sarcoma? One after the other?

At first no. They didn't suppose GIST. They thought that this is leiomyosarcoma with metastasis in liver and in stomach space. After PET CT.

And then they changed their mind and decided. They did a revision and decided that from the beginning it's GIST?

Yes.

They decided that leiomyosarcoma was a mistake?

Well that's also a question to you. Actually as a second opinion.

Okay. So I'll continue reading. So this is the MRI from last August. Where they saw again the liver. Lesions in the stomach. And a 90 centimeter lesion in the uterus. Which seems like a leiomyoma. Not a leiomyosarcoma. And then they saw the revision of the MRI from August. Now when they looked at it. They think that it seems like GIST from the stomach with liver metastasis. A leiomyoma in the uterus. Which doesn't look like uterine sarcoma. And endometrioma in the right ovary. Okay. So let's read the whole letter here of the patient in two parts. Right? So she was diagnosed in September last year with uterine leiomyosarcoma

with liver. Sarcomatosis of abdominal cavity. Greater omentum. And after the fourth. Did you write the summary here?

I don't know what you are reading. But I composed a big document with all the links. Maybe you are reading it.

Okay. After the fourth biopsy change to GIST. In abdominal cavity formation. And GIST metastasis in the liver. And at least the MRI thinks that the uterus and right ovary don't have cancer. Hemorrhagic damage to brain. Wow. After treatment with Pazopanib.

Yes. We did a genetic test in December, OncoMine. And it found the KIT mutation. And again...

Which is very suggestive of GIST, of course.

Yes. But they still thought that this is a sarcoma. And because there was a pathologist's opinion from the Institute of Cancer. And we tried to receive Pazopanib. And there was hemorrhagic damage to the brain. She was hospitalized and recovered from that.

Wow. Okay.

But the treating oncologist later supposed that it could be not PRES. But some sort of tumor lysis. When the tumor was decomposing too fast. But this is just a suggestion. And no one can prove it.

Okay. And then she went on a second line after Trabectedin.

Which was not efficient. Yes.

And I'm trying to understand. She did the PET in April. Did she do a PET before the uptake(?) in the uterus? No, that's the first PET.

No, we did ordinary CTs. PET is not common in Ukraine. We have only a few.

So she just did one PET in April?

Yes.

And then. The uterus and ovary didn't have uptake. But the liver and stomach did have. The liver and the stomach. And the biopsy and revision. And they identified everything as GIST. And then she started the Imatinib.

Yes.

Okay. So she started in, again, August. Tumors in the uterus, ovaries, pelvis, and liver. MRI showed that it was sarcoma. Liver biopsy. Suggested as leiomyosarcoma. The tumor board, the same. Unclear what is

in the ovaries. But according to the MRI, it is not cancer in the ovaries. Okay, so they started Gemcitabine and Docetaxel . Or suggested them. No surgery. No immunochemistry. It was done because they didn't have, okay, that's the first biopsy. And then intoxication psychosis. Wow.

Yes, during the first chemistry, she had intoxication psychosis. And also she recovered from it for some time.

Okay, interesting. The MRI showed a good response in the pelvis, but the liver was worse.

Well, about the response in pelvis. So the Institute of Cancer said that they diagnosed sarcoma because they found small metastases in the great omentum. And they said this is sarcoma's footprint. But I cannot find it anywhere in MRIs and documents. So it goes as anamnesis. But I cannot find where it was originally. But after the first chemotherapy, they said that those small metastases disappeared. So this is, again, sarcoma and the treatment is efficient. But I can't find in the first MRI any mentions of the small metastases. And I gave this MRI to you for review. And your radiologist also didn't mention them.

Right. And then she went on Pazopanib. And because of the KIT mutation in OncoMine?

Yes, a KIT mutation and Met which is subclonal.

And then she had headaches and lost consciousness. And that's what we said about the PRES. Okay. Then they sent a consultation to a sarcoma center in Germany.

It's me. I was being in all the centers.

You sent it.

As well as to you. Well, the sarcoma centers in Germany and others, they didn't suggest anything else. But again, they came out of assumptions that this is sarcoma. And they also consulted me as for sarcoma.

Who did a revision of the slides, all the pathology slides? Did the center in Germany see the slides?

No, they didn't request the pathology material, the biopsy. I sent you the very first biopsy as well. But the problem is that this is the first sample, it's already small and many Institutions were taking from it. So it's almost nothing left...

But she had more biopsies later on?

Yes. She had more biopsies, but they were unsuccessful, for example the biopsy which was taken in January, it had RNA destroyed. And I sent it to Germany, but for Foundation testing... Genetic testing, but unfortunately, they couldn't extract RNA. Also I sent it to you and you also said that there was a mistake during the preparation of it. So the only working biopsies are the very first one and as one taken in April.

Which weren't good enough to give a pathological diagnosis?

Well, I gave it to you for review.

So we have to discuss it because I'm not a pathologist. And then she started the Imatinib.

Yes.

Just a month ago and how was she feeling with the Imatinib?

15:25 Well, she felt immediately better when she started taking Imatinib. But maybe she felt better also because she wasn't receiving chemistry anymore... chemotherapy. But right now she says that she's still weak in the second part of the day and she doesn't feel she's recovering further. Yes, but this huge expansion of the liver which caused pains because, as they explained, the liver was great and created pressure on the capsule. It is gone right now. So...

Okay, so she started the Imatinib, it's better.

Yes.

16:04 I think according to what I hear that she didn't have a leiomyosarcoma. It was all a mistake from the beginning and it was GIST from the beginning and really Imatinib is the only right drug until now because we know that Pazopanib isn't the first line and Docetaxel/Gemzar also don't work in GIST. And Imatinib is the best drug and usually it does work quite fast. And she started only a month ago. So it's still early to do imaging. But I hear from you that she feels better with the Imatinib and it's probably not just because she's not getting chemotherapy because it's probably at least the right drug. Also the KIT mutation supports this and also the revision of the MRI which tells us and also the PET that probably what they have... what they see in the uterus isn't a cancer at all. So I think She's getting the right treatment and in another two months she can do a CT and reassess the liver. And I want to try and open, let's see... The OncoMine genetic analysis... Let's see. I'm looking at all the attachments which are very nice. But first of all, I want to see the... And let's see if I can open it. Okay, so what they see here and I'm gonna call up for filing.

19:11 You open the sequencer files, yes? With DNA and...

19:13 The results. So the mutation in KIT in 42% of the cells. Which has sensitivity to KIT inhibitors like the Imatinib she's getting. She has other translocations which is interesting and other mutations the tumor mutational burden is low. The micro satellite is low, which is what we usually see in a GIST. But I want to see... How common the TPMT mutation is in GIST.

20:13 Well, I have provided the files from the sequencer such so it's one file for DNA, one for RNA. They said that DNA can be read by anyone and RNA for them because this is their own primers library.

20:31 Mm-hmm. I saw. I'm looking in a CBO portal. I'm opening up a database of 25,000 tumours which were sequenced... all kinds of tumours. Not just GIST. Okay, so It's not found in any kind of cancer, which is interesting. Let's see what you know about it. So it's a genetic polymorphism for chemotherapy but it's not connected to cancer. So we have to put it aside and then we have the KIT mutation.

I don't understand. What does it mean? So what do you mean? There is no cancer?

22:21 The TPMT? No, no. No, I mean that in the mutations there is... the opto mutations in TPMT. But this gene isn't connected to cancer, it's connected to metabolism of chemotherapy. So it doesn't give us any ideas how to treat it so specifically, does it or it doesn't help us know what is the origin of the tumour. If I look specifically at the KIT mutation... So the KIT mutation is very specific to GIST. 80% of all GIST have a KIT mutation also in non-melanoma skin cancer and melanoma and germs and tumour in endometrial cancer. It is very very rare, about 4%. So it also supports the idea that it's not uterine leiomyosarcoma.

23:26 But no one actually supposed at that moment that this is a GIST because they said that the leiomyosarcoma also might have a KIT mutation.

23:36 So that's what I told you now that It's very very very rare that uterine leiomyosarcoma have a KIT mutation. That's what I opened. I can, I'll show you.

23:50 Okay, I see.

23:52 Do you see my... Let's see, if you can see my screen... one moment. Do you see the graph here? I'm sure you can see it so much.

I see the upper part. You can share...

Here. This is better.

Mm-hmm

24:20 Okay, so this is a site called CBO portal where they have it's the largest site which is open to the public and Genomic testing of different tumours and I asked him from 25,000 tumours that were and they said genomic testing how many... which kinds of tumours had any mutation in GIST. So that this large Bar here shows that it's GIST. So from all the 25,000 tumours 900 of them which is 3% had a KIT mutation - any KIT mutation - and close to 80% of them were GIST. All the other ones are very very very low. It's like there's much much more common in GIST than anything else. Now I want to look at that specific mutation to see if they have information about not any KIT mutation, but the specific KIT mutation the little sister has in the tumour so... I look...

25:41 One question, you say that something is this genetics is due to chemotherapy. But the genetics was made from the very first sample before...

25:54 No, not that it's due to chemotherapy, that it affects the response to chemotherapy, but not the kind that she got. So it doesn't matter at all. And... So this is what she has an extinct, let's check. And to understand, I'm looking for any information about the specific mutation. So if you open the OncoMine, so it says glycine five five six... so glycine is... No, it's not, literally it's glutamine, it's Q, and VAL is valine, which is V. So if I go back here, she has a mutation with a Q five five six and V five five nine. So Q five five six V five five nine deletion which is this. Okay? Do you see my...

27:27 Well, I barely see...

It's hard to see.

27:32 Yes. I see that you selected some point but I do not understand genetics up this extent. So what does it mean? You may explain...

27:43 Okay. So from all what I'm saying is that from, you know, the KIT is a gene and it has many many kinds of mutations inside the same gene. So I'm looking for the specific protein that changed according to the OncoMine results to be very very specific, not any KIT mutation, but a specific KIT mutation. And so I see that three tumors in this database had the same mutation where the Q, like they wrote here, the GLN, which is Q 556, changed to valine, which is V 559. And all three mutations which were described in this database of 25,000 tumors, all three of them were in GIST. So we can be very very sure that this is a GIST and that it and like you see in the recommendations that Imatinib is the first line. So you need a second time to go to Afatinib(??)... pretty pretty these are the recommended lines. So now she's getting the first line, which is Imatinib. I'm not sure you can see it. If you want to... so really we know that a KIT is much more common in GIST but even her specific mutation was only described in GIST... now if you want to ask me... okay, but, but... If we look at the database not of all the cancers, but if you we really really want to check specifically for

uterine cancers so I can check, I can choose here only uterine sarcoma. Well, they have it's a very rare cancer so it's not endometrial carcinoma and it's not uterine carcinosarcoma, but I chose here uterine sarcoma database of 108 samples and here I want to ask him if that is the database if there is any KIT in uterine leiomyosarcoma and a zero zero of the cases of uterine sarcoma in this database said cKIT. So I can really say that also according to the OncoMine results It's not uterine leiomyosarcoma also according to the MRI and the PET. So we can be sure... that, here... they when they did a revision here of the biopsy so they did do the staining for C-KIT and it's SDH negative, which is interesting. Also the liver was the same and the SDH deficient is a kind a very specific kind of GISTs which usually begins in the stomach... five percent of the GIST are SDH deficient like her type of tumor and it's more common in young people who have GIST. And they also have very very, like... It doesn't respond well to chemotherapy, okay, all the GIST don't respond well to chemotherapy, but I want to check - in very rare cases, it's genetic. Very very rare. Did anybody in your family have cancer besides her?

32:02 Well, in my family, there is a big history of cancers so my Granny from the mother's side. So the sister is my sister from the mother's side. So we have different fathers, but we have the same mother and actually we have cancers on the mother line so granny had cancer of the liver and mother had four years ago double cancer of uterus and breast there's also different types of cancer and she had surgery removing uterus and cancer in the breast.

So I really suggest genetic testing...

For BRCA, yes? But...

32:53 Not just BRCA because BRCA isn't connected to GIST. Like... You know, today we do genetic testing for 150 genes at one one blood test. They check all the genes that can be connected to anything. So like wide-range generally easy to do today, it's much cheaper than OncoMine but the price and they check all. Looking at... It's interesting because when they say that... Because according to the literature, if there is a mutation in... There is supposed to be also a deficiency in SDH, and SDH is... It's deficient....to tell us it has an inhibitor, so the Imatinib shouldn't work if it's SDH-deficient, like they wrote here. On the other hand, first of all, she is responding clinically, and also she has a mutation in KIT, so I'm not so sure about this SDH deficiency. They understand what I said because of two reasons because the SDH shouldn't come with a mutation in KIT and also shouldn't respond to Imatinib. So let's leave the SDH an open question but of course that genomic testing is recommended for the full panel, and not just BRCA. So I think...

34:32 Please, excuse me, before we go, can you write that in your conclusion?

Yes, of course.

So you propose... So you say that there are two mutations in GIST that contradict each other, and she may not respond well to Imatinib, yes?

34:52 I think If we have to say which one is right. So I would believe more the OncoMine that showed the mutation in KIT because also the pathology shows staining for KIT and in all three specimens also the abdominal cavity also the liver and the second liver which is interesting and... You know, the tumor can always be heterogenic, some parts like this, or like that, but I would say that she already started the Imatinib and she is responding so let's leave the SDH for a moment and believe more the C-KIT also the staining, also the mutation go together, or much more common. It's like 80% is data and SDH deficiency is 5% so I wouldn't change treatment because of this SDH deficiency, because it's not going well with the whole clinical picture, so I don't know about it.

36:11 What you recommend to do, so the question is, if it will be beneficial to do some additional genetic tests like foundation, so the doctors told me that generally it will be, so it's not needed right now, because it's just, and it's being treated, but when we were treating sarcoma, there was a question that there can be about 5% chance to find something else, when a genetic test is done by different equipment and different programs.

36:46 That is right, that is right, the different companies that do genomic testing of the tumor are supposed to give us all the same results, but we know that even in the same specimen, different companies give different results, like you're saying, the equipment is different, the bioinformatic analysis of the data is different, they use different databases, or different criteria for what is a pathogenic mutation and what is just a normal variant, so it's a good question. I think that 99% it would be, you would get exactly the same results, it's a very specific mutation in C-KIT, it's a rare mutation, like we saw, it's only 3% out of the samples, out of the 90 samples that had any KIT mutation in this database, of the 25,000 tumors, so it's really, really rare, but I think that you would just get exactly the same result, if you do it with a foundation, so I'm not sure you have to do it now, but it's an interesting question, because it is a very rare mutation.

38:17 But if the Imatinib is not working, you have another suggestion for this mutation of a drug?

38:27 If the imatinib isn't working, I would try again to do a new biopsy, and understand better, and then look again at the pathology, again check the SDH to see if it's real or not, again check the mutations... it's an interesting case... but I think that right now, I really hope that finally she's getting the right treatment.

38:53 Okay. So, you've already said that conclusion, so I far understood that there's a contradiction between Oncomine and your own pathologist, so your pathologist finds SDH deficiency, while Oncomine doesn't have it?

39:07 They can't look for it, because it's not a mutation, it's a staining, they didn't look for it at all. The contradiction is more between the SDH and the C-KIT staining, which aren't supposed to come together, but it's not always according to the book.

Okay.

Okay, what else do you want to ask?

Well, so I composed a number of questions for my contacts of the clinic, I don't know whether you have received them. So I can ask...

Questions? No, I went over everything you sent. Just before the questions, I just want to go over the side effects that you should follow for the Imatinib. So, 400 mg a day is the full dose, which is okay. The side effects that can happen are edema around the eyes, I don't know if she has it.

No, she doesn't have any visible side effects right now.

Okay, so edema around the eyes, in the hands, in the feet, you have to follow the blood tests, also complete blood count, biochemistry, liver, kidneys, thyroid function, blood pressure, it can cause sometimes a bit of diarrhea, a bit of nausea, but usually if she's a... I would mainly follow... I didn't see any blood tests, I don't know how her liver function tests were before she started.

I have all the blood tests collected in my...

It's in the file.

In the... at the end of the... if you look in the Google documents...

I'm looking, yes, yes, yes.

At the very end, there are all the blood tests which were made...

Ah, I see, I see, I see.

And you can check the last blood tests.

41:17 Great, wow, you're very, very efficient. Okay, let's see, from the 15th of May, okay, great. So, the ALT and AST are close to normal, also the amylase is just a bit high, the LDH isn't too high, which is good, alkaline phosphatase is high, we would want to see it going down.

It's not very, very high, but it is. The bilirubin is high, which I think... Ah, no, it's not the same units as in Israel. No, according to the units, it's okay, it's not like here. I'm sorry, here the units are different for the bilirubin. The total protein is okay, which is a very good sign, the creatinine is also okay. Okay, not bad, the blood count right now is okay. The electrolytes are also... Okay, so not bad. Let's open, let's see if I can... She started it in the beginning of May, right?

Yes.

Imatinib. So, I just want to have a look at the blood tests of the liver before she started. Okay, so the ALT, AST were okay, the LDH was also okay, alkaline phosphatase just a bit. Okay, the bilirubin went down, but it was okay, according to the creatinine. Okay, so not too bad. I got it in all four. Listening to your questions.

Let me see, so I can ask my questions, yes?

Yes, yes, I'm listening.

Okay, so the question is, so we suppose that there is no any malignant tumor in the uterus and ovaries, so...

Right. Also, that there isn't now, and also that there wasn't any in August, according to what it seems now.

43:44 So, actually, on the very... the biopsy, the Institute of Cancer, they didn't do immunochemistry, they concluded the tumor is malignant of mesenchymal nature, and they suppose that this is a leiomyosarcoma, but as far as I understood, GIST is also similar to sarcoma.

44.04 Exactly. GIST is also a type of sarcoma.

Yes.

I understand that the biopsy, the first biopsy wasn't taken from the uterus, right? It was from the abdominal cavity?

Yes, it was from the liver, but the patient remembers that they took it from the different side of the liver. She supposes that all the biopsies were taken from the one side of the liver, because there was a huge metastasis, which was selected as a control one. But...

But she never had a biopsy from the uterus or from the ovary?

No, because Institute of Cancer feared that they will cause strong metastasis if they poke through the shell of the tissue. So, they had...

44:57 Okay, did she ever have a gastroscopy done? To look for the primary in the stomach?

Yes, she had a gastroscopy. So, when she was... In April, when she was taking the biopsy from... When she was taking the biopsy from GIST, they used the equipment to look through the stomach. So, they told me that they will search how can they take this biopsy, either from inside of the stomach or from outside. And after the examination, they decided to go from outside. So, and they did what they call the thin needle biopsy, and they used it as far as I understood, the camera, ultrasound camera, which they inserted into the stomach. So, as far as I understood, they couldn't reach this GIST from the stomach. They used external needle. This is what I know.

And that's the pathology that they couldn't use? They couldn't look at?

What do you mean?

And so, they took it through the gastroscopy, but from... They went with a fine needle aspiration through the stomach wall?

Let me see if there is a document after that. Let me enter the drive.

I want to... I'm trying to understand if they saw the tumor there or not. If they saw the tumor in stomach?

Yes.

It seems to me... Well, I don't know. So, all that I can know is written in their documents. Let me... Let me see. There should be documents of the... Maybe I should have left that after that. Let me see. So, everything is in that folder. But let me see if there is something.

So, I'll look again at that you sent. Maybe I'm wrong? No, that's again the fourth biopsy in April from the abdominal cavity and the liver was GIST, but I'm looking for the... Well, what is it?

47:40 Probably, they did the gastroscopy. They did that in place and decided how to take the biopsy and they did not create a conclusion. Let me see. There's a document. No, so they didn't issue any documents. So, there are only those pathohistological studies from April and May, from 21st of April. And that's it. So, they didn't tell me. So, they didn't create a conclusion from the gastroscopy. So, the only thing that I know is that they planned initially to take the samples through the stomach and they wanted to use special equipment for that. But after that, at the place, they changed their opinion into doing a thin needle biopsy from the external, from outside.

49:27 So, I would... I would like to make a suggestion that I think that endoscopic ultrasound with the biopsy is something that she should do sometime just to have the primary tumor sample and check.

But one of the biopsy is a primary tumor, I suppose, because they... So, this is a tumor from... So, they use a thin needle biopsy to take the sample from this tissue near the stomach. So, 21st...

We don't know, but we don't have a result of the pathology from this.

We have. So, look, in the document, Overview.docx, there are two pathologists. So, you can see a pathological histological study "21st of April" and pathohistological study "26th of April". And 21st of April is... So, let me see. It's... There are two samples taken. One from...

I'm looking. I'm opening it. One moment. Let's see. So, the 21st of April.

Is this a specimen from near the stomach?

But it's the abdominal cavity. It's not a gastric wall. The abdominal cavity we know that...

So, yes. They took from this formation in the abdominal cavity, which they suggest to be the primary tumor. And the second one is 26th. It's from metastasis.

I see.

And yes, they wanted to take it through the stomach, but during the examination, they changed to a thin needle from the outside. Is that...

51:26 Okay, interesting. Because the primary of the GIST is supposed to be right inside the muscle of the gastric wall and not through the wall into the abdominal cavity. So, like I would think that sometime along the way, the endoscopic ultrasound would be an interesting test to do to see. Inside the gastric wall, not outside in the abdominal cavity. Because the GIST probably started in the stomach. So, she had a lot of biopsies, so we don't have to do it now, but just sometime along the way.

Yes. So, you suggest to research how it looks from inside the stomach and to make a conclusion.

Yes, endoscopic ultrasound.

Endoscopic Ultrasound, okay. It will be very good if you write that in conclusion, because I can make those examinations and then get back to you, for example, with them.

Great. So, consider an endoscopic ultrasound.

Just you can leave some short commentary for me to explain that to doctors, why it should be made. So, because GIST is originated in the stomach and we need to know what happened here. Maybe they just didn't have the right equipment, because I first needed to do that in one clinic, which is supposed to be professional and that they call it gastro clinic and so on. But the doctor wasn't available and I changed it to an oncology clinic. And they said that they can do this, but it will be slightly different. So, I don't know what did they mean, it's slightly different. Maybe they didn't have the needle to poke through the stomach. Maybe they had only the camera and the external needle. So, maybe they just didn't have that equipment.

53:47 Okay. Okay. I want to write here the recommendation. One moment. Okay, so it's suggested to continue Imatinib, PET-CT in 3 months(?), see if there is a response(?), ultrasound examination of a tumor in stomach(?), (?) not expanding, consider biopsy for SDH deficiency and cKIT. Okay, what else? [Not clearly heard because of echo]

No, I just want you to write the recommendations for the doctors here, because many of them may not even know what we are talking about. So, you can write, please, about the CSDH deficiencies, so I can ask them and they may do some examinations, which may help. So, please write about this question. So, I suppose they know what does it mean in terms of Imatinib, that it might be inefficient. Because she will undergo the check in the middle of June. And I can ask the doctor what does she think and to do some additional examinations. Maybe another pathology study.

56:19 Okay. I'm writing here also genetic testing for germline patients. A whole panel. Not just the BRCA, but everything. Also the SDH and everything.

Okay.

56:38 Okay. If you do... not the Foundation, Foundation don't do genetic testing for germline mutations. But usually here in Israel, we send to Tempus, which do, do have, when they do genomic testing and the tumor, they also do germline testing. They do somatic and germline together. So, if you sometimes think about repeating the genomic testing of the tumor, so I would suggest doing it with a company that does also germline and also somatic mutations.

Can you somehow, we are talking right now, but the only thing which is left is your conclusion. I may forget something.

Yes.

Can you please write down, and write down this company, maybe I can reach them in Israel. And maybe I can send it to them. Maybe your clinic can send the samples to them.

57:35 You have to ask the Manor people, what they say. Okay?

Okay.

57:42 I wrote it down. Consider Tempus for both tests together, germline and somatic. They wouldn't check, like the SDH. I write it down, specifically SDH. Just to make sure. That if it's not in the panel, so they can add whatever we want.

Thank you very much.

You're welcome. I'll write my email if you have questions or updates.

I have another question, actually.

Okay. So, then the next question is, so she has leiomyoma, which is not malignant. But I read that it can transform into leiomyosarcoma. And also on the very first MRI, it was active, it was accumulating contrast. So the question is how dangerous it is, and what can we do with it? So to make it not to turn into cancer.

58:53 I think she has to do a PET another two months. And if the PET is quiet, so just wait until the next PET. I wrote it. I'll add it to the suggestions. Follow up for uterine findings, which are considered benign. But right now, but should be followed up. Should be followed up.

What do you mean? So what can be done with it?

With the PET. Just the PET. When she does the PET, to see how she is responding to the Imatinib. So they will also look at the uterus.

But what can be our actions if we see that it evolves? It does something. It's doing something.

59:48 Take a biopsy. Take a biopsy through a gynecological ultrasound, which is very easy to do. And see if it's a... GIST shouldn't go to the uterus? Like it's... And to think that she has something else in the uterus. It's also like anything can be, but it's not very something.

Yes, but I don't know how to understand for the future. So if it's dangerous, if it can mutate. And right now, as far as the surgical operations are not welcome. Because the organism will be weakened during them. But can we somehow probably remove it, for example, from the inside of the uterus? Maybe they can...

01:00:30 No, no, no, I don't think there is any reason to do it. Because she already has the GIST. And we aren't worried that something different will develop in the uterus. If something has happened, so we will check it then. But right now, we are only worried about what we have now, which is that...

So if it starts... Sorry for the harsh question. If it starts to mutate into leiomyosarcoma, then we will start treating leiomyosarcoma?

Yes, but the risk of it is very, very, very low.

Yes, but...

Leiomyoma usually doesn't mutate into leiomyosarcoma. I'm more worried about the GIST. The liver. That's the main issue now.

01:01:27 Okay, I see. Okay, so also I have the question. So when she received the first protocol, the Docetaxel and...

Gemcitabine. Gemzar.

Gemcitabine. Yes. There was a very huge reaction. She had psychosis intoxication and she had a huge process in the liver. The liver expanded very much and 80% of it was various cysts... There is an ultrasound conclusion in the docs and they even attached the image from ultrasound. And also there is an image of the liver in...

I saw.

Yes. So the question for me is why the liver reacted so much if there was no oncology for this chemistry? So... And metastasis actually stayed more or less the same in size. So what it was?

01:02:17 Side effect of the chemotherapy. The GIST doesn't respond to chemotherapy. It's not chemotherapy sensitive. So it's... And we know that the Taxotere sometimes does do neurological side effects. Very rarely, but it can happen. So it sounds to me that the tumor didn't get the right treatment so it was not effective. And on the other hand, it caused side effects.

So you think that all this expansion of liver, it's not actually oncological. It's a side effect from...

No, it can also be from the tumor that didn't get the right treatment.

Okay, I see. Okay, and how do you think... What do you think, how can we recover the liver? So can the actual liver be recovered? Can Imatinib destroy those metastases at all? Or how can we recover the liver to get...

01:03:26 The Imatinib, if it works, it will make the metastasis smaller, much smaller, I hope. Much less active in face(?). She will feel better. But Imatinib is the only way to make the liver better.

Okay. Okay, so then the question. So as far as I understood, the Imatinib or any other target will make the metastasis and the main GIST to shrink. But it will still remain. Can we somehow remove the GIST completely? So maybe some non-invasive methods like gamma knife or freezing.

01:04:06 It's a very good question. Right now, we want to see the Imatinib is working. If for another six months or nine months, everything is gone by the Imatinib. Except some very small lesions, so we can think about something local. But right now, it's too big for local measures. And really Imatinib is the best thing for her. And probably along the whole way, Imatinib will be the best thing for her. So usually in GIST, we don't do anything local unless there is a special response.

So you say that if the response to Imatinib is very good and the tumor shrinks very much, then we can do something to destroy the tumor.

Maybe. Maybe.

Okay. But actually, the oncologist already warned me that the tissue adapts to the treatment after some time and like in three to five years or something...

Right.

To change to the next one.

01:05:18 Right. But first of all, we have a long time until then. And all the time, we have new research for new drugs that come out, which are getting better and better, there are other inhibitors like Imatinib. So I think that if she will respond well to the Imatinib, and she will need a change another three to five years. So probably there will have new drugs that we don't even know about today. Because the research is continuing all the time.

Okay. So the question is, what's the quality of life? So she wonders that she still feels not very good. Is it possible that she can get back to work as an IT specialist? Can she feel more or less okay? So maybe when the tumor shrinks enough. So is her condition due to the liver or due to the cancer itself? And so...

It's the same. The liver is because of the cancer. It's the whole issue together. Once she starts responding...

I don't hear you. Something happened to the sound. You're muted.

I'm sorry. Do you hear me now?

Yes.

And I have patients who came like with metastatic GIST and responded very well and went back to work. And it does... Three to five years is a very long time in oncology for responding to first line. And we have right now four lines of treatment, which probably means she will... If she responds, she will have a very long time to enjoy life, to enjoy work, whatever she likes. Yes. If there's a good chance, she will feel much better and go back to whatever she likes.

So she can work, she can do some sports like walking in the forest, etc.

I hope, yes. Yes, I hope so.

So when GIST... So her processes in the organism will be mostly normal when the GIST will shrink.

I hope so, yes.

So her feelings will get more to the feelings of a normal person.

Right.

Yes, because she's concerned that she will be like handicapped from the rest of her life. So she will be like...

No, no, no, no. She will feel better and she will be okay.

Okay. So, and my last question we covered mostly. So which research examinations and which tests we should undergo? So the question was about foundation. You suggested foundation is not the case right now. Is it better cases, the tests you proposed?

Right.

Okay.

I wrote down the tempus. Tempus, because they also do the germline testing in the same time. And it really can add a lot of information.

The problem is, which I face, is that many doctors in Ukraine, they do not work with such tests as they don't know how to comment on them. So...

But you can speak to...

To your clinic, yes?

Yes, yes, you can. After we have the results, we can make an appointment again and discuss it. Yes.

Okay. I think that I learned from this conversation a lot. Thank you very much. If I have some other questions, or maybe if my treating oncologist will make some suggestions, I can... Can I write to your email about them?

Yes, yes, yes. I printed it out, everything.

Yes, I can pay for any additional consulting. Just I want to keep in touch in case some questions appear.

Great. Great, whatever you want.

Yes. Okay, thank you very much in this case.

Okay, you're welcome. So I hope your sister feels well. And I think you're doing a great job, doing everything you can to help her.

Thank you.

Which is very, very nice. Okay, so I'll see you.

Okay, thank you. Bye.

Bye. You're welcome.

Bye.

Bye.